

Cryptography Lab Report

Experiment 01: Caesar Cipher

Caesar Cipher is a substitution cipher where each letter in the plaintext is shifted by a fixed number of positions in the alphabet.

Encryption: $C = (P + K) \bmod 26$ | Decryption: $P = (C - K) \bmod 26$

Experiment 02: Vigenere Cipher

Vigenere Cipher is a polyalphabetic substitution cipher that uses a repeating keyword to determine the shift for each letter.

Encryption: $C = (P + K) \bmod 26$ | Decryption: $P = (C - K) \bmod 26$

Experiment 03: Playfair Cipher

Playfair Cipher encrypts pairs of letters using a 5x5 matrix constructed from a keyword. It improves security over simple substitution.

Experiment 04: Hill Cipher

Hill Cipher uses matrix multiplication to encrypt blocks of text. The key is an invertible matrix modulo 26.

Encryption: $C = (K \times P) \bmod 26$

Experiment 05: Rail Fence Cipher

Rail Fence Cipher is a transposition cipher that writes text in a zigzag pattern across multiple rails and reads row-wise.

Experiment 06: RSA Algorithm

RSA is an asymmetric cryptographic algorithm using public and private keys. It is based on the difficulty of factoring large numbers.

Experiment 07: Diffie-Hellman Key Exchange

Diffie-Hellman allows two parties to generate a shared secret key over an insecure channel using modular exponentiation.